Introduction

Skiplist and Multilist: Data Structures and its Applications (UE25CS52A)

N S Kumar
Professor
Department of Computer Science & Engineering (AI & ML)

© Confidential Information of PES University. All Rights Reserved - 2026

Quiz : Skiplist and Multilist

1 What is true in a skiplist?

a) There are multilevels, each of which has the same number of nodes
b) The number of pointers in each node depends on the number of levels planned
c) The traversal for any operation starts at the higher level and this level has all the nodes
d) Any insertion will affect atmost one level

© Confidential Information of PES University. All Rights Reserved - 2026

Quiz : Skiplist and Multilist

1 What is true in a skiplist?

a) There are multilevels, each of which has the same number of nodes

b) The number of pointers in each node depends on the number of levels planned

c) The traversal for any operation starts at the higher level and this level has all the nodes

d) Any insertion will affect atmost one level

© Confidential Information of PES University. All Rights Reserved - 2026

Quiz : Skiplist and Multilist

2 What is true in a multilist?

a) This is same as a multidimensional array
b) Every node will have only single pointer
c) Every node may participate in multiple lists
d) Addition of a node would require updation in only one list

© Confidential Information of PES University. All Rights Reserved - 2026

Quiz : Skiplist and Multilist

2 What is true in a multilist?

a) This is same as a multidimensional array

b) Every node will have only single pointer

c) Every node may participate in multiple lists

d) Addition of a node would require updation in only one list

© Confidential Information of PES University. All Rights Reserved - 2026

Quiz : Skiplist and Multilist

3 sparse matrix is an example of

a) Using more space to gain in time
b) Using less space at the cost of time
c) a 2D array which can be directly indexed
d) A matrix where most of the elements are non zero

© Confidential Information of PES University. All Rights Reserved - 2026

Quiz : Skiplist and Multilist

3 sparse matrix is an example of

a) Using more space to gain in time

b) Using less space at the cost of time

c) a 2D array which can be directly indexed

d) A matrix where most of the elements are non zero

© Confidential Information of PES University. All Rights Reserved - 2026

Quiz : Skiplist and Multilist

4 In a SkipList, the bottom-most level contains:

a) Only the largest element

b) Only the smallest element

c) All the elements in sorted order

d) Randomly selected elements

© Confidential Information of PES University. All Rights Reserved - 2026

Quiz : Skiplist and Multilist

4 In a SkipList, the bottom-most level contains:

a) Only the largest element

b) Only the smallest element

c) All the elements in sorted order

d) Randomly selected elements

© Confidential Information of PES University. All Rights Reserved - 2026

Quiz : Skiplist and Multilist

5. A multilist is particularly useful when:

a) Data needs to be accessed through multiple relationships

b) Only sequential access is required

c) Memory must not contain pointers

d) Only one ordering of data is needed

© Confidential Information of PES University. All Rights Reserved - 2026

Quiz : Skiplist and Multilist

5. A multilist is particularly useful when:

a) Data needs to be accessed through multiple relationships

b) Only sequential access is required

c) Memory must not contain pointers

d) Only one ordering of data is needed

© Confidential Information of PES University. All Rights Reserved - 2026

Skiplist and Multilist

ADT: Revision

- Lists and stacks can be implemented in different ways while supporting the same interface.
- Separating concerns leads to good software development.
- A data structure has components; each component should have its own behaviour.
- Example: a polynomial can be represented using an array or a list of terms.

© Confidential Information of PES University. All Rights Reserved - 2026

Skiplist and Multilist

Skiplist: Intuition

- A normal shuttle bus stops at every village and may take a long time.
- An express bus can skip villages and then a local bus can reach the destination.
- This idea of skipping portions of a sorted list is called a skip list.

© Confidential Information of PES University. All Rights Reserved - 2026

Skiplist and Multilist

Limitation of a Normal List

- Example: 10 → 20 → 30 → 40 → 50 → 60 → 70 → 80 → 90 → 100
- Searching for 90 requires traversing almost the whole list.
- Worst-case steps and comparisons are proportional to the number of elements

© Confidential Information of PES University. All Rights Reserved - 2026

Skiplist and Multilist

Structure of a Skiplist

- Level 2: 10 ——————→ 40 ——————→ 70 ———→ 100
- Level 1: 10 ——→ 20 ——→ 40 ——→ 50 ——→ 70 ——→ 100
- Level 0: 10 → 20 → 30 → 40 → 50 → 60 → 70 → 80 → 90 → 100
- Higher levels provide shortcuts;
- level 0 walks through the list one step at a time.

© Confidential Information of PES University. All Rights Reserved - 2026

Skiplist and Multilist

Searching for 90 in a Skiplist

- Start at 10 on the highest level.
- Move to 40, then 70 because each is less than 90.
- The next node is 100, which is greater than 90.
- Drop down one level at 70.
- Drop one more level at 70.
- At level 0: 70 → 80 → 90; reach 90

© Confidential Information of PES University. All Rights Reserved - 2026

Skiplist and Multilist

Searching for 90 in a Skiplist

- Start at 10 on the highest level.
- Move to 40, then 70 because each is less than 90.
- The next node is 100, which is greater than 90.
- Drop down a level at 70.
- At level 0: 70 → 80 → 90; reach 90

© Confidential Information of PES University. All Rights Reserved - 2026

Skiplist and Multilist

Why the Skiplist Helps?

- Higher levels allow us to jump over large portions of the list.
- Instead of examining all eight elements before 90, only a few nodes are examined.
- A skip list is a sorted linked list augmented with multiple levels of forward pointers.

© Confidential Information of PES University. All Rights Reserved - 2026

Skiplist: Structure and Operations

- A skip list may have any number of levels.
- The number of levels depends on the expected number of nodes.
- Normally, each higher level has about half the nodes of the next level.
- Each node has active pointers for the levels in which it participates. Other pointers are grounded.
- Insertion and deletion follow the same top-down idea as searching.

© Confidential Information of PES University. All Rights Reserved - 2026

Skiplist and Multilist

Skiplist: Applications and Salient Points

- In-memory ordered data for faster insertion, deletion and searching.
- Concurrent data structures where operations can happen in parallel.
- Priority ordered events.
- Can be used instead of balanced search trees.
- Bottom level contains all elements; higher levels contain fewer elements and provide shortcuts.
- Note: Implementation of a skiplist is not part of the source's stated syllabus.

© Confidential Information of PES University. All Rights Reserved - 2026

Skiplist and Multilist

Multilists

- A multilist means a list of lists.
- A node may be part of more than one list.
- A list may have nodes whose keys are heads of another list.
- The same node can be reached through different linked lists representing different relationships.

© Confidential Information of PES University. All Rights Reserved - 2026

Sparse Matrix: Motivation

- Very large matrices may contain mostly zero elements.
- Storing every element consumes a lot of memory.
- Store only non-zero elements together with their row and column information.
- For efficient rowwise and columnwise traversal, maintain a row list for each row and a column list for each column.

© Confidential Information of PES University. All Rights Reserved - 2026

Skiplist and Multilist

One Physical Node, Multiple Logical Views

- Example: students take courses, and each course has students.
- One view finds the courses taken by a student.
- Another view finds the students who have taken a course.
- Keeping separate structures would duplicate data and require multiple updates.
- A multilist captures the mapping and supports both traversals.

© Confidential Information of PES University. All Rights Reserved - 2026

Skiplist and Multilist

Applications of Multilists

- Sparse matrix
- Graph representation
- Social networks
- Multiple views as in files

© Confidential Information of PES University. All Rights Reserved - 2026

Multilist: Salient Points

- Supports Multiple relations
- Avoids duplicates
- Avoids multiple updation
- Has Complicated pointer management
- Has Complicated insertion and deletion

© Confidential Information of PES University. All Rights Reserved - 2026

Skiplist and Multilist

Summary

- Skiplist: a sorted linked list with multiple levels of forward pointers.
- Higher skiplist levels provide shortcuts for faster searching.
- Multilist: a list of lists in which a physical node can participate in multiple logical lists.
- Multilists are useful when the same data needs multiple views or relationships.

© Confidential Information of PES University. All Rights Reserved - 2026

THANK YOU

N S Kumar
Professor

Department of Computer Science & Engineering (AI & ML)